

Methodological Trade-offs in QML

A Comparative Analysis of La Wun and Arjun's Pipelines

Executive Summary: This document maps the theoretical and empirical differences between two robust approaches to the CICIOT2023 dataset. La Wun's pipeline anchors on operational realism, tackling complex multi-class hardness with empirical bounds. Arjun's pipeline is architected around mathematical certainty, focusing on subset analytic certificates. Rather than competing paradigms, they function as dual lenses: one mapping real-world operational friction, the other establishing upper-bound theoretical guarantees.

Day 13: MAQT Hyperparameter Tuning (λ_1, λ_2)

While both pipelines utilize a unified loss function combining cross-entropy with geometric regularizers, their tuning methodologies diverge significantly.

RATIONALE & DESIGN

La Wun: Evaluates across a complex, full-multiclass environment (0.5% subsample). The driving philosophy is downstream resilience, strictly optimizing for the validation macro-F1 score under a fixed λ constraint throughout all epochs. The environment is deliberately harsh to simulate actual dataset constraints.

Arjun: Operates on a balanced 6-class subset, prioritizing Hilbert geometry (trace distance and fidelity). A curriculum warmup is employed, meaning the hyperparameters act as a steady-state target after an initial ramp-up. The goal is geometric stability rather than immediate classification prowess.

RESULTS & SYNTHESIS

La Wun's broad grid search suggests $\lambda_1 \approx 0.66$, acknowledging that λ_2 has a weaker immediate impact on F1. Arjun locks in $\lambda_1 = 0.5, \lambda_2 = 0.02$ based on trace distance tightness. The team's eventual shared default ($\lambda_1 = 0.5, \lambda_2 = 0.3$) is an elegant compromise, anchoring on Arjun's mathematically stable λ_1 while adopting a moderate λ_2 that nods to La Wun's empirical observations.

Day 15: Proposition 1 / Global Depolarizing Channel Λ_p

Both pipelines successfully validate the identity $D_{tr}(\Lambda_p(\rho), \Lambda_p(\sigma)) = (1-p)D_{tr}(\rho, \sigma)$ with machine-precision accuracy, though their notebook architectures serve different functions.

RATIONALE & DESIGN

La Wun: Treats Day 15 as an intensive, standalone study. The verification sweeps across varying Hilbert dimensions ($n \in \{1, 2, 3\}$ qubits) and explicitly ties the theoretical channel to a NISQ circuit demo and regularizer contraction curves. The approach is deeply pedagogical and engineering-focused.

Arjun: Implements Day 15 as a streamlined checkpoint within a massive end-to-end multi-day notebook. The verification tests solely at the production scale (8-qubit, dim 256) to establish the necessary $(1-p)$ factor for later steps, prioritizing execution speed and pipeline continuity.

Day 16: Lipschitz Constant (L_φ) Estimation

The calculation of the Lipschitz constant defines how sensitivity is modeled, contrasting mathematical bounds with statistical realities.

RATIONALE & DESIGN

La Wun: Publishes an empirical $p95$ estimate ($L_\varphi \approx 2.92$) derived directly from data pairs and probes on the trained encoder. The philosophy dictates that operational limits should reflect the actual distribution of gradients observed in the wild.

Arjun: Deploys Analytic Lemma 1, providing a hard certified bound ($L_\varphi \leq 1.5$) based on structural assumptions of a 3-layer architecture. This establishes a theoretical ceiling immune to sampling variance.

Trade-off: Arjun's analytic bound is smaller and cleaner for subsequent proofs. La Wun's empirical constant is larger, enforcing a more conservative downstream radius. It acts as a necessary diagnostic check against overly optimistic theoretical assumptions.

Day 17: Certified Radius (R)

Using the identical formula $R = m / (2(1-p)L_\varphi C_\rho)$, the two pipelines yield vastly different radius distributions.

RATIONALE & DESIGN

La Wun: Evaluates R across nearly 19,000 full-multiclass test points without conformal rejection filters. It is a raw measure of the classifier's margins in a hostile, uncurated data environment.

Arjun: Measures R on a 4,000-sample top-6 subset, explicitly coupling the calculation with conformal calibration to analyze only accepted samples ($s \leq q$).

RESULTS & SYNTHESIS

Arjun reports a mean $R \approx 0.093$, benefiting from the smaller analytic L_ϕ and an easier classification task. La Wun reports a conservative mean $R \approx 0.008$. Together, they illustrate the spectrum of model capability: Arjun maps the theoretical potential in a controlled environment, while La Wun maps the absolute worst-case operational baseline.

Day 18: Proposition-2 Geometry (ϵ^*)

The assessment of the separable budget definitively unites the two pipelines under a shared conclusion, despite differing definitions of in-class fidelity (F_{in}).

RATIONALE & DESIGN

La Wun: Defines F_{in} by the minimum own-class fidelity, a strict constraint that pulls the geometric difference (Δ) heavily negative.

Arjun: Aligns with the theoretical assumption (A2), defining F_{in} as the minimum F_{max} over the known test set.

RESULTS & SYNTHESIS

Both methods yield negative values for the separation metric ($\Delta < 0$) and the budget ($\epsilon^* \approx -0.151$ for La Wun; -0.105 for Arjun). This mutual failure of Assumption 5 is a powerful scientific consensus: it proves that the sufficient condition provided by Proposition 2 is too conservative to guarantee a positive L_2 budget on the CICIOT dataset. By arriving at the identical functional conclusion from opposite ends of the empirical-theoretical spectrum, both pipelines validate the necessity of moving beyond worst-case assumptions toward subsequent empirical breakpoint analyses.